

Expt. No. : 5

Date :

FILTER EVEN NUMBERS FROM A LIST

AIM

To write a Python program to separate even number from the given list of Numbers.

SOFTWARE REQUIRED

Python 3.14

FORMULA USED

Even Numbers:

n is even if $n \% 2 == 0$

$n \% 2$ calculates the remainder when n is divided by 2. If the remainder is 0, n is even.

THEORY

An integer is classified as even if it is divisible by 2 without any remainder. Mathematically, a number n is even if there exists an integer k such that:

$$n = 2K$$

Examples of even numbers include 2, 4, 6, 8, 10, etc

PROBLEM

1. Build a Python program to separate the even and odd numbers from a given list and display them in two different lists.

2. An electrical utility company maintains the ratings of distribution transformers. Build a Python program that accepts a list of transformer ratings and returns a new list containing only the transformers with ratings greater than the specified capacity.

3. Consider a list (`list = []`). You can perform the following commands:

`insert 'i' 'e'`: Insert integer 'e' at position 'i'.

`print`: Print the list.

`remove 'e'`: Delete the first occurrence of integer 'e'.

`append 'e'`: Insert integer 'e' at the end of the list.

`sort`: Sort the list.

`pop`: Pop the last element from the list.

`reverse`: Reverse the list.

Initialize your list and read in the value of 'n' followed by 'n' lines of commands where each command will be of the 7 types listed above. Iterate through each command in order and perform the corresponding operation on your list.

ALGORITHM

STEP 1: Start the program.

STEP 2: Receive a list of numbers as input

STEP 3: Create an empty list to store the even number.

STEP 4: Use a loop to go through each element in the input list.

STEP 5: For each element, check if it is an even number (divisible by 2)

STEP 6: Return the list containing only the even numbers as the result

STEP 7: Terminate the program.

PROGRAM CODING

```

even = []
odd = []
input_1 = []
n = int(input("Enter how many numbers: "))
for i in range(n):
    num = int(input("Enter number: "))
    input_1.append(num)
    if num % 2 == 0:
        even.append(num)
    else:
        odd.append(num)
print("Elements in the List:", input_1)
print("Even Numbers:", even)
print("Odd Numbers:", odd)

```

OUTCOMES OF PROGRAM CODING

```

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experiment5.py =====
Enter how many numbers: 5
Enter number: 54
Enter number: 87
Enter number: 56
Enter number: 41
Enter number: 23
Elements in the List: [54, 87, 56, 41, 23]
Even Numbers: [54, 56]
Odd Numbers: [87, 41, 23]

```

PROBLEM CODING

```

ratings = []
n = int(input("Enter how many transformer ratings: "))
for i in range(n):
    rating = int(input("Enter transformer rating: "))
    ratings.append(rating)
capacity = int(input("Enter maximum capacity: "))
result = []
for i in ratings:
    if i > capacity:
        result.append(i)
print("Transformers greater than capacity:", result)

```

OUTCOMES OF PROGRAM CODING

```

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File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on
win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment5.py =====
Enter how many transformer ratings: 5
Enter transformer rating: 110
Enter transformer rating: 220
Enter transformer rating: 230
Enter transformer rating: 400
Enter transformer rating: 415
Enter maximum capacity: 230
Given transformer ratings: [110, 220, 230, 400, 415]
Transformers greater than capacity: [400, 415]
>>>

```

PROBLEM CODING

```

lst = []
n = int(input("Enter number of commands: "))
for _ in range(n):
    command = input().split()
    match command[0]:
        case "insert":
            lst.insert(int(command[1]), int(command[2]))
        case "print":
            print(lst)
        case "remove":
            lst.remove(int(command[1]))
        case "append":
            lst.append(int(command[1]))
        case "sort":
            lst.sort()
        case "pop":
            lst.pop()
        case "reverse":
            lst.reverse()

```

OUTCOMES OF PROGRAM CODING

```

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=====RESTART: C:\Users\Mugil\Desktop\Python\Problem1\5.py=====
Enter number of commands: 6
append 10
append 20
insert 1 15
print
[10, 15, 20]
reverse
print
[20, 15, 10]

```

VIVA QUESTION

1. What is a break, continue, and pass in Python?

- break: Exits the current loop entirely when encountered.
- continue: Skips the rest of the current iteration and moves to the next iteration of the loop.
- pass: Does nothing and is used as a placeholder where code is syntactically required but no action is needed.

2. What are Built-in data types in Python?

Python has several built-in data types, including:

- Numeric types: int, float, complex
- Sequence types: list, tuple, range
- Text type: str
- Set types: set, frozenset
- Mapping type: dict
- Boolean type: bool

Binary types: bytes, bytearray, memoryview

3. How do you floor a number in Python?

To floor a number (round it down to the nearest integer), you can use the `math.floor()` function from the `math` module.

4. What is the difference between xrange and range functions?

In Python 2, `range` returns a list, while `xrange` returns an iterator that generates values on demand, which is more memory-efficient. In Python 3, `xrange` was removed, and `range` behaves like `xrange` in Python 2, returning an iterator.

5. What is the purpose of the len() function in Python?

The `len()` function is used to return the length (i.e., the number of elements) of an object such as a list, string, tuple, or other collection types.

RESULT:

Thus, the Python program successfully accepted a list of numbers from the user, filtered the even numbers, and returned them as a new list. The program executed correctly, and the output was verified successfully for the given test cases.

INFERENCE:

The experiment demonstrates that Python can efficiently process lists and filter elements based on a specified condition. The successful execution and verification of the program confirm the correctness of list operations, conditional statements, and user-defined functions in extracting even numbers from a given list.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		